

Network Topology

Lecture 9

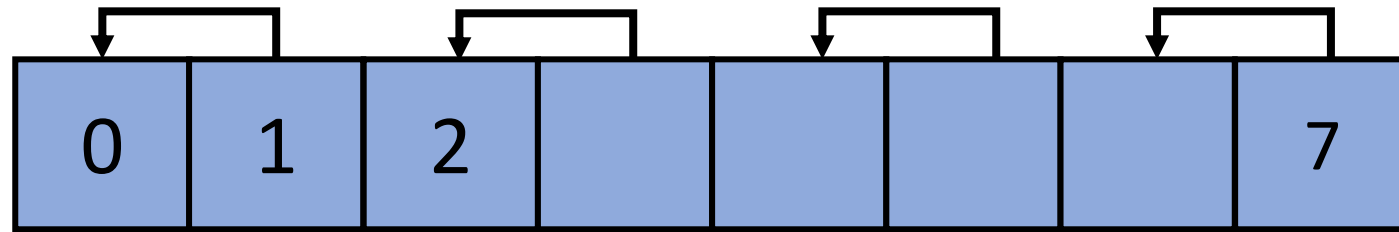
February 4, 2026

Timing – Doubt (from G forms)

Report memory allocation time
(anything that is done once
before iterative communication
steps) as initialization time

```
if (option == 1)
    //standard communication mode
    MPI_Send(arr, BUFSIZE, MPI_INT, size-1, myrank, MPI_COMM_WORLD);
else if (option == 2) {
    buf = (char *)malloc( bufsize );
    MPI_Buffer_attach(buf, bufsize);
    //buffered communication mode
    MPI_Bsend(arr, BUFSIZE, MPI_INT, size-1, myrank, MPI_COMM_WORLD);
}
else if (option == 3)
    //synchronous communication mode
    MPI_Ssend(arr, BUFSIZE, MPI_INT, size-1, myrank, MPI_COMM_WORLD);
```

Recap: Virtual Topology and Communication Graph



Process decomposition will determine the communication pattern and the virtual topology

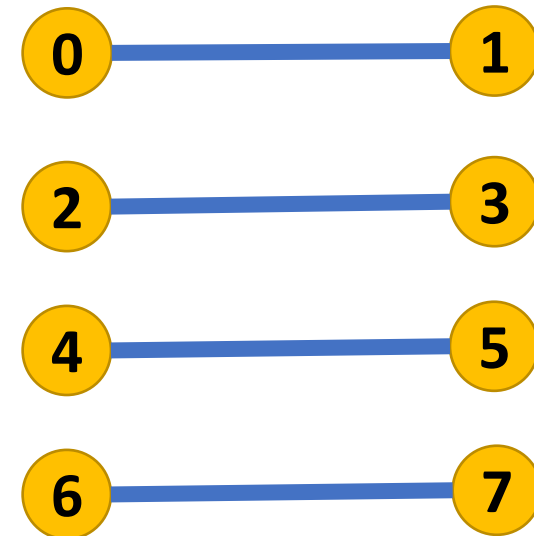
$1 \rightarrow 0$

$3 \rightarrow 2$

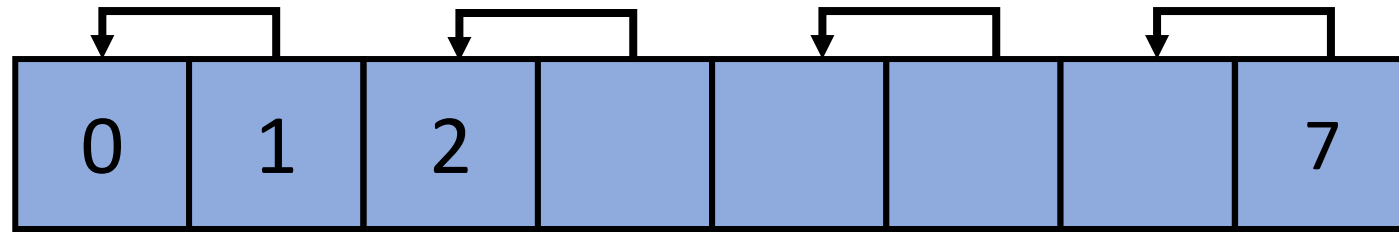
$5 \rightarrow 4$

$7 \rightarrow 6$

Note this is the communication pattern for the above process decomposition



Recap: Process Placement/Mapping



Number of ranks = 8, Nodes requested = 2, ranks/processes per node (ppn)=4

Rank 0 is placed on Node 100

Rank 4 is placed on Node 102

Rank 1 is placed on Node 100

Rank 5 is placed on Node 102

Rank 2 is placed on Node 100

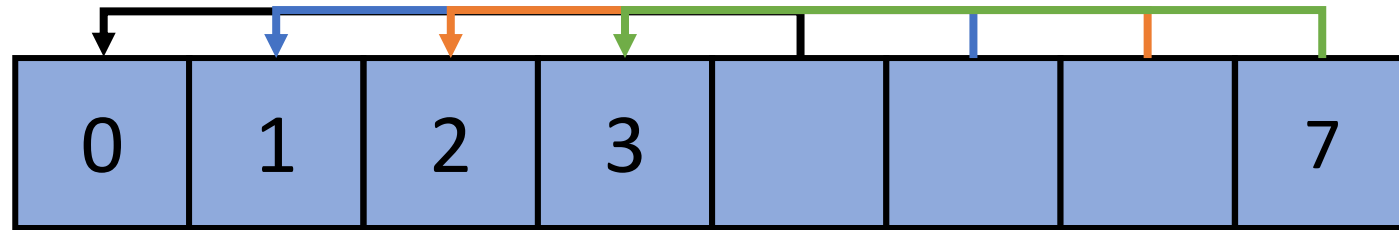
Rank 6 is placed on Node 102

Rank 3 is placed on Node 100

Rank 7 is placed on Node 102

This is the default node-level mapping

Process Placement/Mapping



Number of ranks = 8, Nodes requested = 2, ranks/processes per node (ppn)=4

Rank 0 is placed on Node 100

Rank 4 is placed on Node 102

Rank 1 is placed on Node 100

Rank 5 is placed on Node 102

Rank 2 is placed on Node 100

Rank 6 is placed on Node 102

Rank 3 is placed on Node 100

Rank 7 is placed on Node 102

Can we do better than the default mapping?

Yes, for e.g.: 0, 4, 1, 5 may be placed on Node 100 and 2, 6, 3, 7 on Node 102

Attributes/Characteristics of Interconnects

GGKK Chapter 2

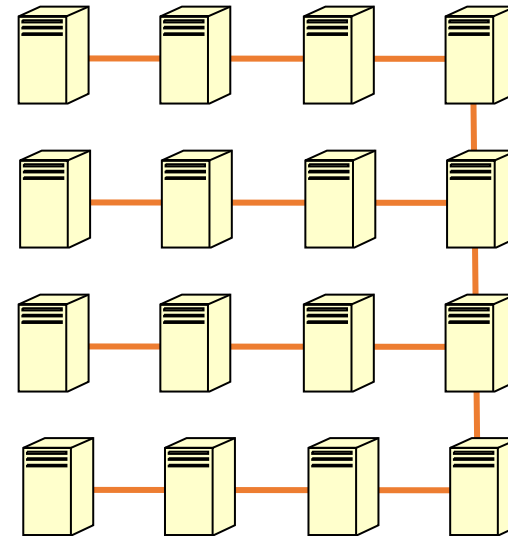
- Topology
- Diameter
- Cost
- Bisection width – The smallest number of edges that have to be removed from a graph/network to partition it in two halves [1]
- Bisection bandwidth – The sum of bandwidths of the minimal number of edges that have to be removed from a graph/network to partition it in two halves (roughly two equal parts)

[1] Bisection (Band)Width of Product Networks with Application to Data Centers, IEEE TPDS 2014 (Optional reading)

Hypothetical Network

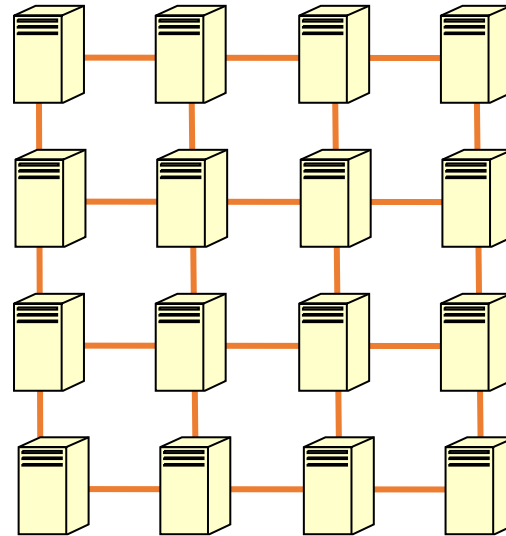
Total #nodes = p

- Diameter $3(\sqrt{p} - 1)$
- Bisection width 1
- Cost $\sqrt{p}(\sqrt{p} - 1) + (\sqrt{p} - 1)$



2D Mesh Interconnect ($\sqrt{p} \times \sqrt{p}$)

Total #nodes = p



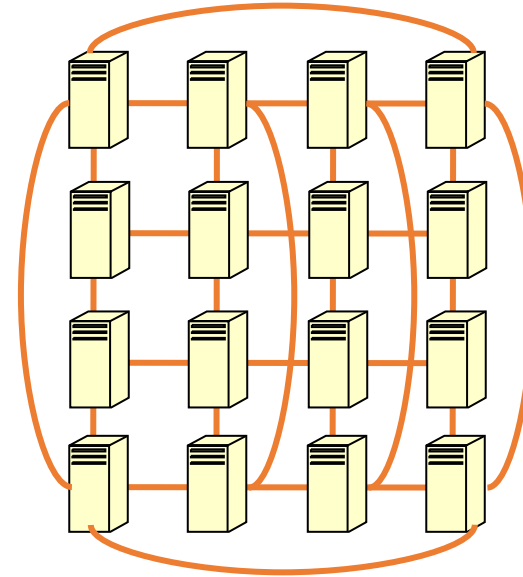
- Diameter $2(\sqrt{p} - 1)$
- Bisection width $\sqrt{p}$
- Cost $2(p - \sqrt{p})$
 $2\sqrt{p}(\sqrt{p} - 1)$

Simple improvement
over this interconnect?

2D Torus Interconnect ($\sqrt{p} \times \sqrt{p}$)

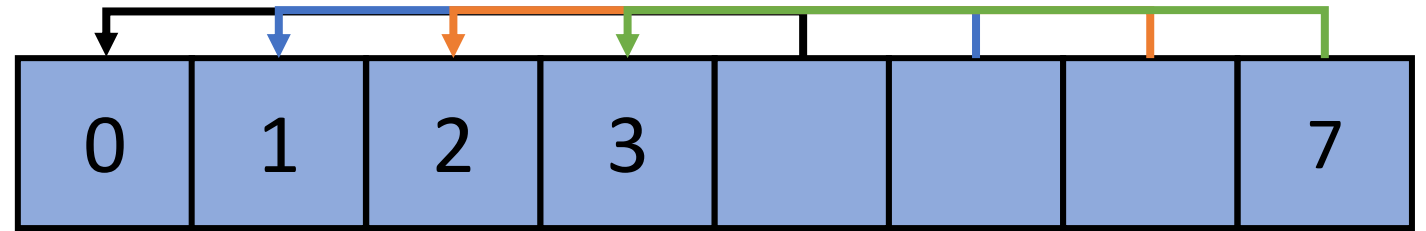
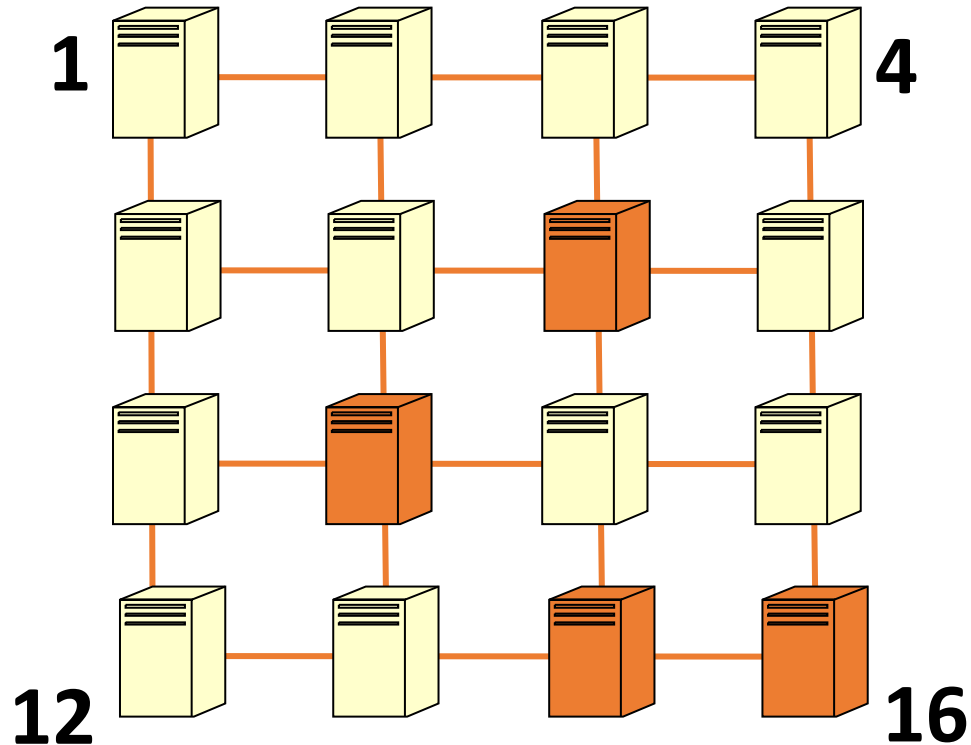
Total #nodes = p

- Diameter $\sqrt{p}$ [$=2(\sqrt{p}/2)$] (when p is even)
 $\sqrt{p} - 1$ (when p is odd)
- Bisection width $2\sqrt{p}$
- Cost $2p$
 $2\sqrt{p}(\sqrt{p} - 1) + 2\sqrt{p}$



Note: All network links are not shown to avoid visual clutter

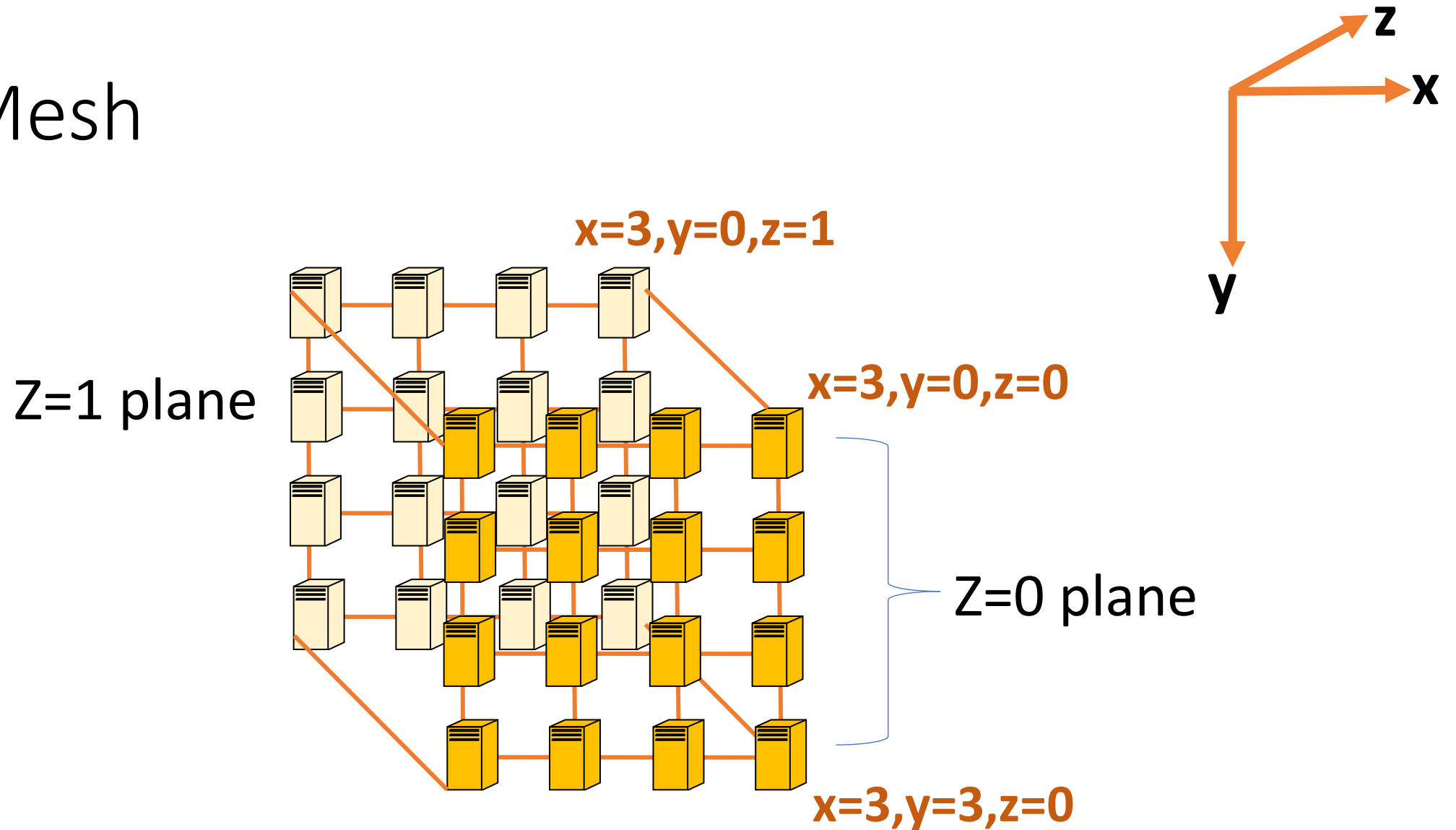
In-class Assignment



Number of ranks = 8, Nodes requested = 4,
ranks/processes per node (ppn)=2

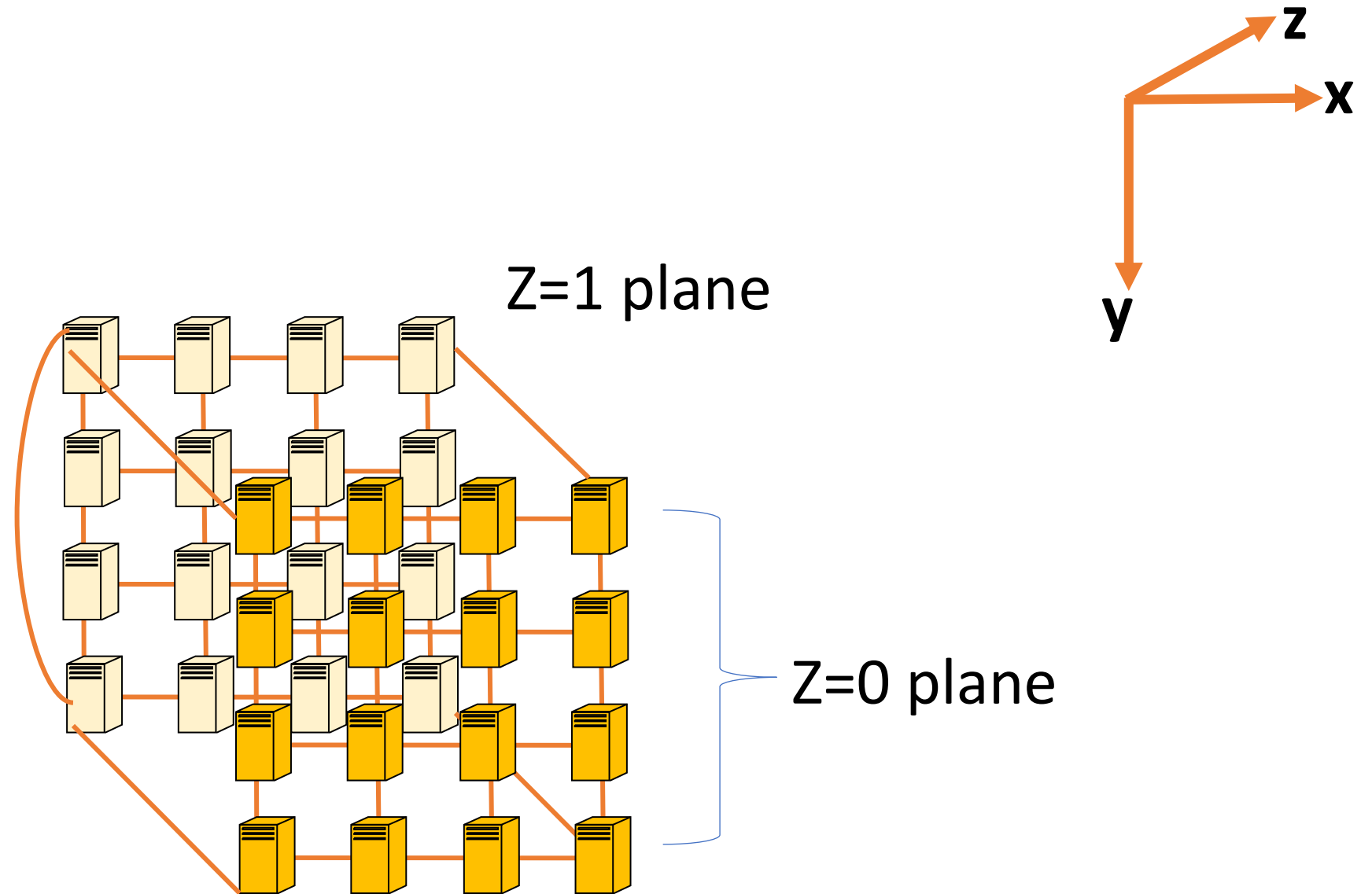
- Draw the communication graph
- Maximum number of hops?
- Average number of hops?

3D Mesh



Note: All network links are not shown to avoid visual clutter, different colours are chosen for nodes/links for visual clarity

3D Torus



Note: All network links are not shown to avoid visual clutter, different colours are chosen for nodes/links for visual clarity

Homework – Derive the network parameters

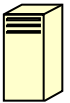
- 3D Mesh ($v_p \times v_p \times v_p$)
- 3D Torus ($v_p \times v_p \times v_p$)

Hypercube Interconnect

2^4
4D

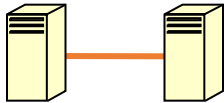
#Nodes (p) 2^0

0D



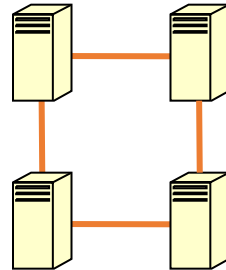
2^1

1D



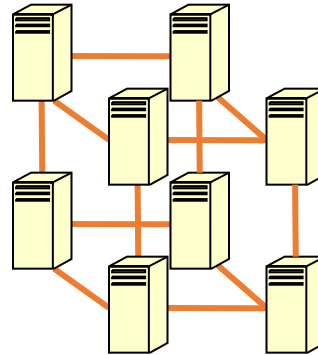
2^2

2D



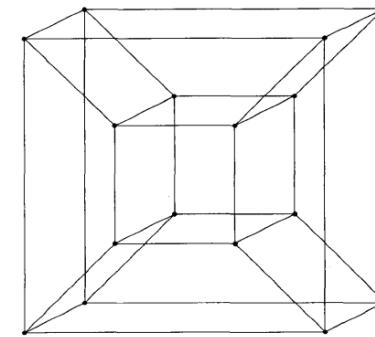
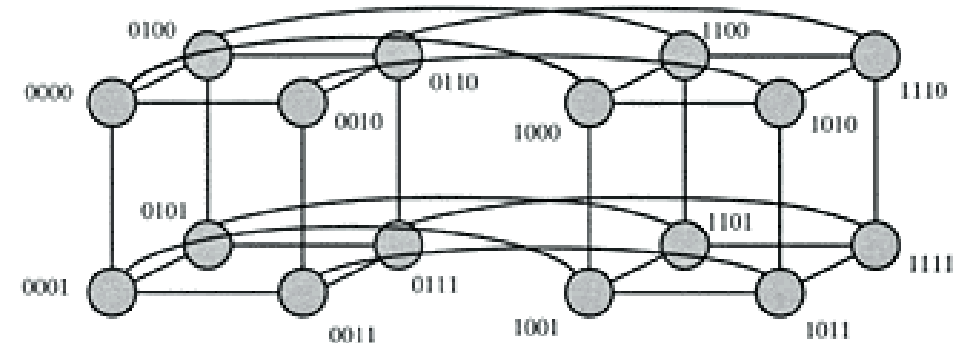
2^3

3D



Source

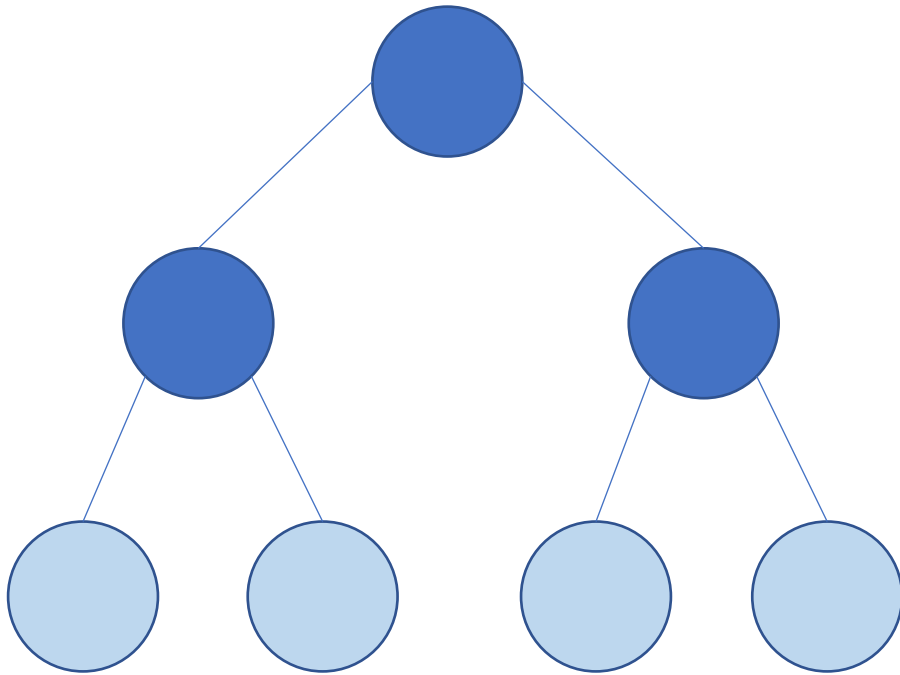
https://www.researchgate.net/publication/220693596_Parallel_Algorithms_for_Regular_Architectures_Meshes_and_Pyramids



- Diameter $\log p$
- Bisection width $p/2$
- Cost (Homework)

Source: Saad and Schultz, Topological properties of Hypercubes, [IEEE Transactions on Computers](#)

Tree Networks



Problem?

Communication bottlenecks at higher levels

Fat-tree Networks

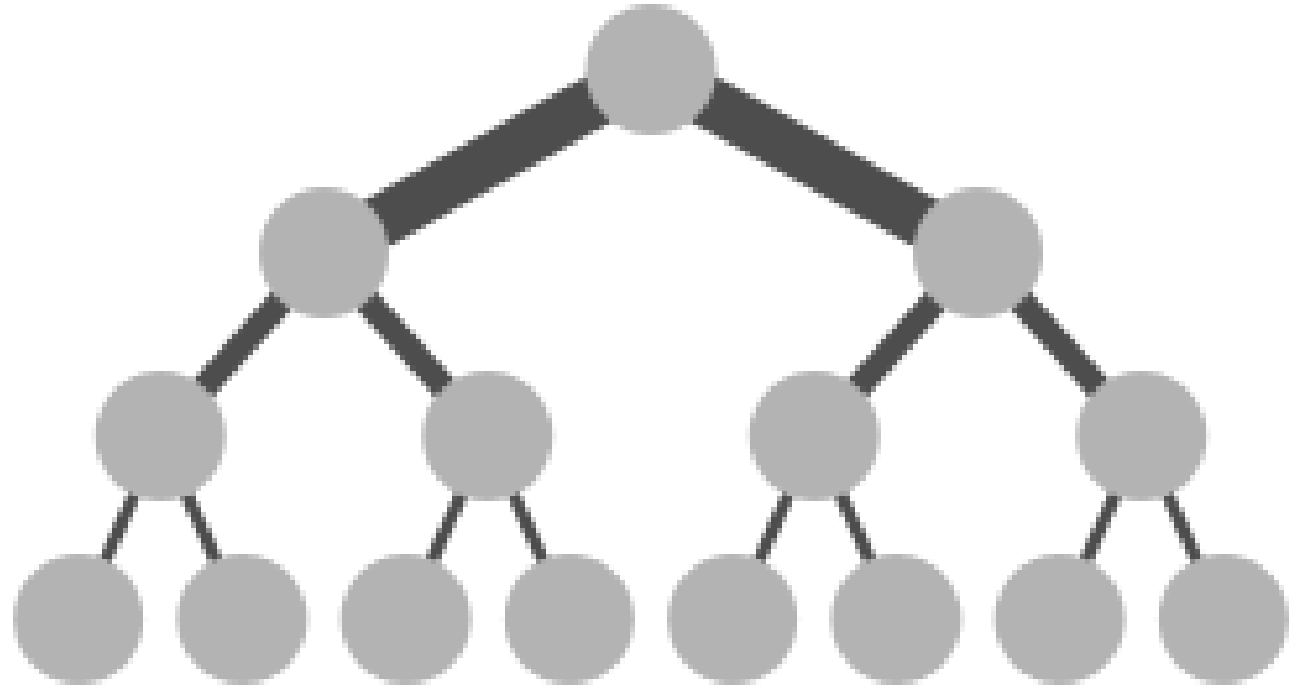
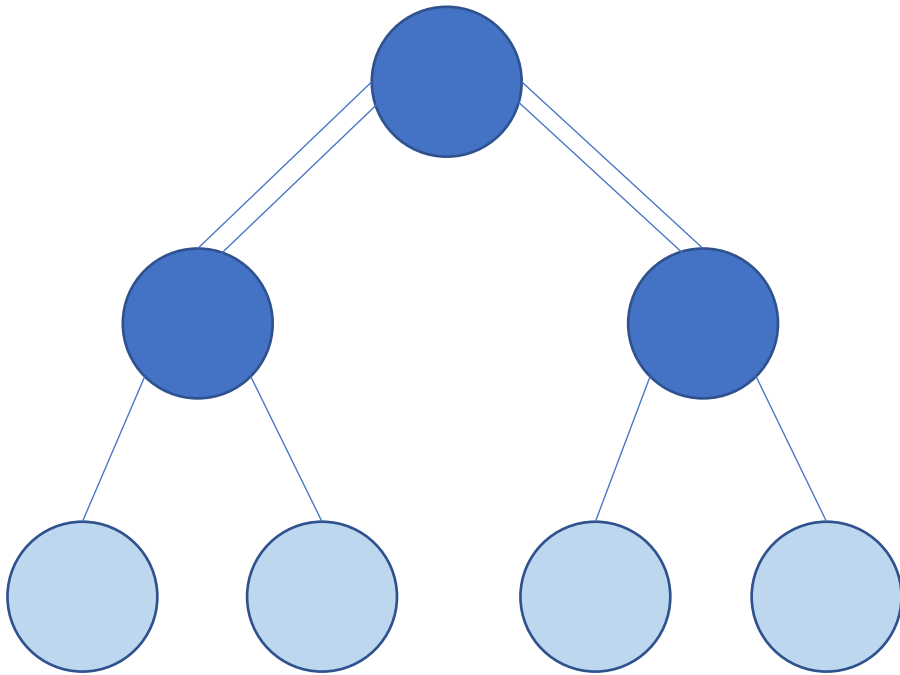
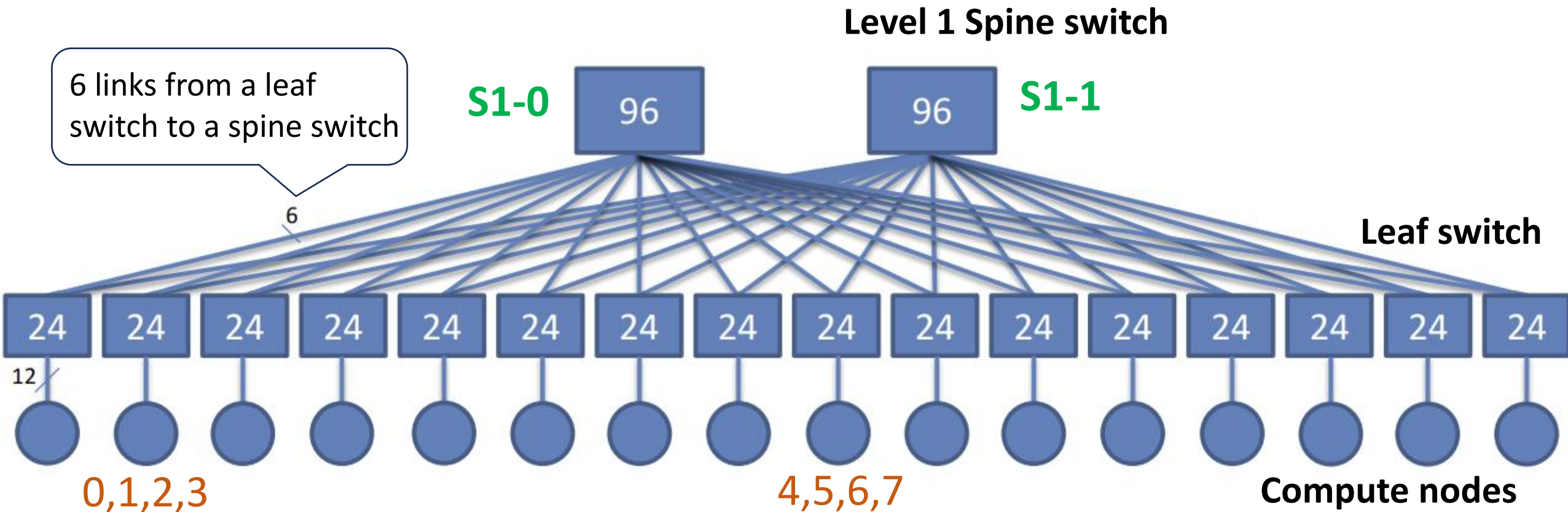


Image source: Wikipedia

Optional reading: C. E. Leiserson. Fat-trees: Universal networks for hardware efficient supercomputing. In *Proceedings of the 1985 International Conference on Parallel Processing*, 393–402, 1985.

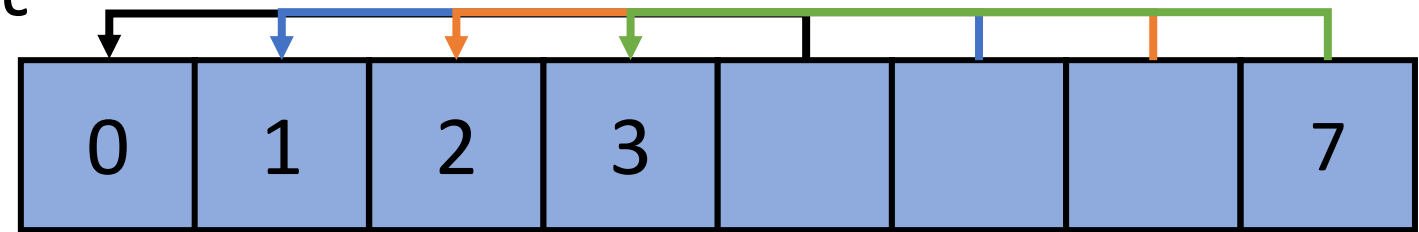
2-level Fat-tree Network



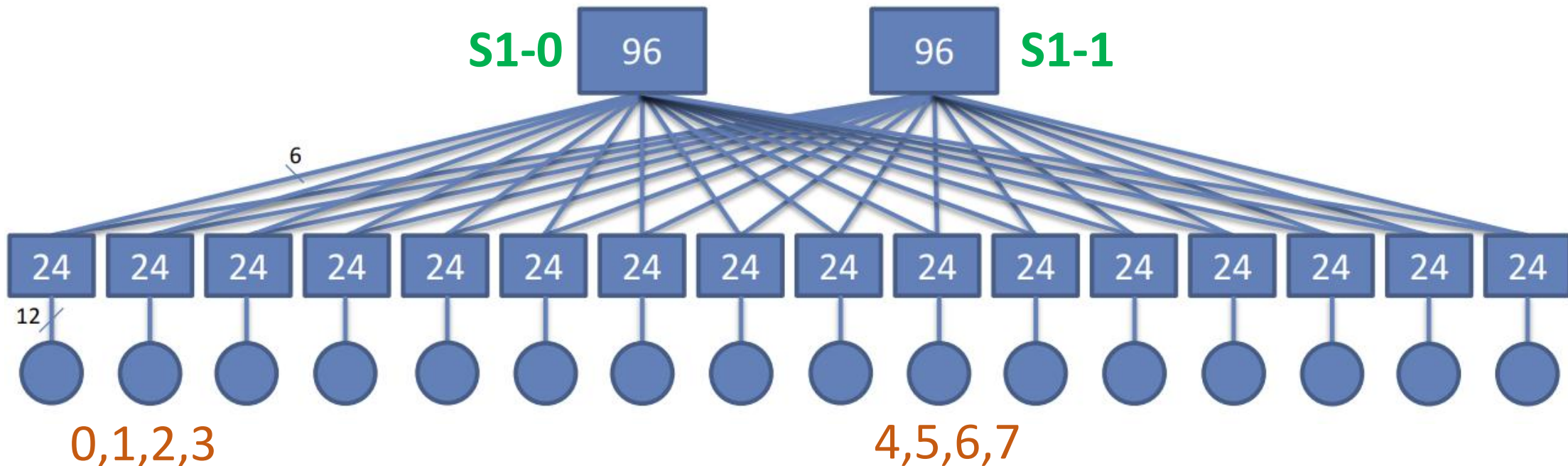
Bisection width: 96 [Remove half the links, each from S1-0 and S1-1]

In-class Assignment

- Maximum number of hops?
- Average number of hops?



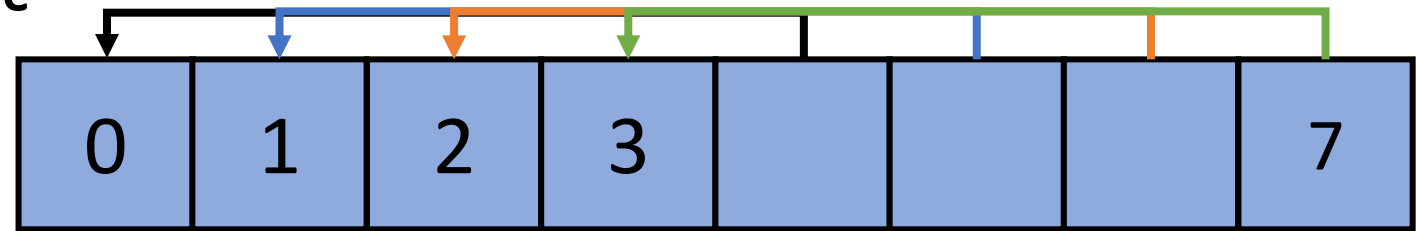
Number of ranks = 8, Nodes requested = 4,
ranks/processes per node (ppn)=2



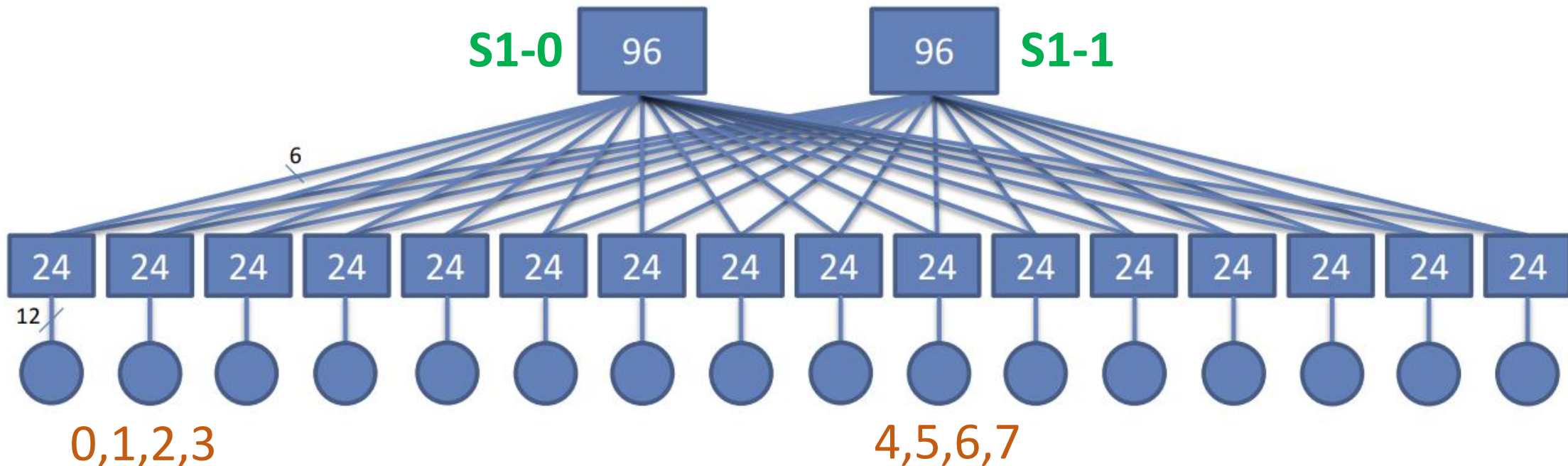
In-class Assignment

Q: Will there be congestion (assume no other job is running)?

A: Not on the leaf to spine switches, only on the compute node to leaf switch



Number of ranks = 8, Nodes requested = 4,
ranks/processes per node (ppn)=2



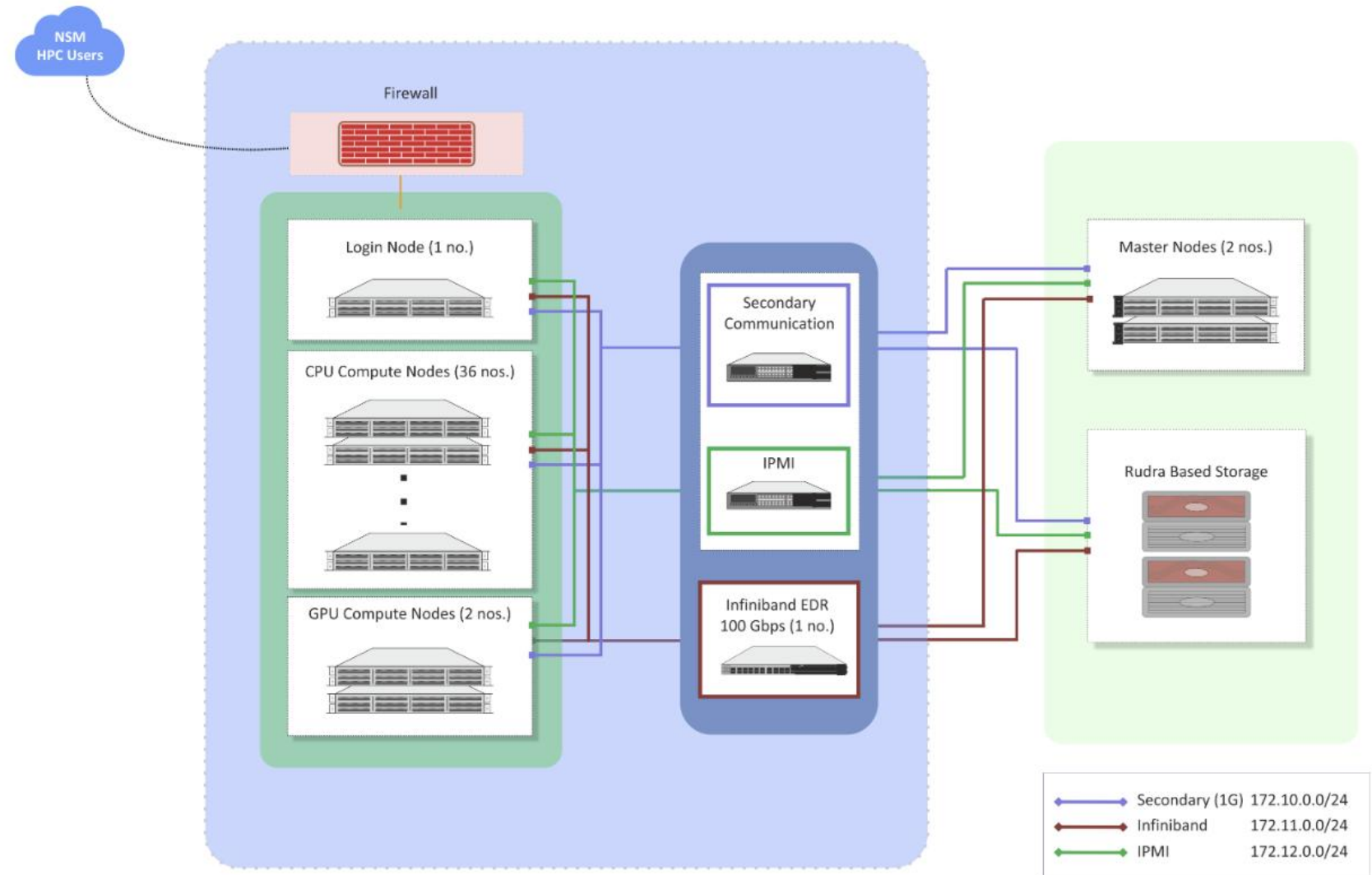
Papers (Optional Reading)

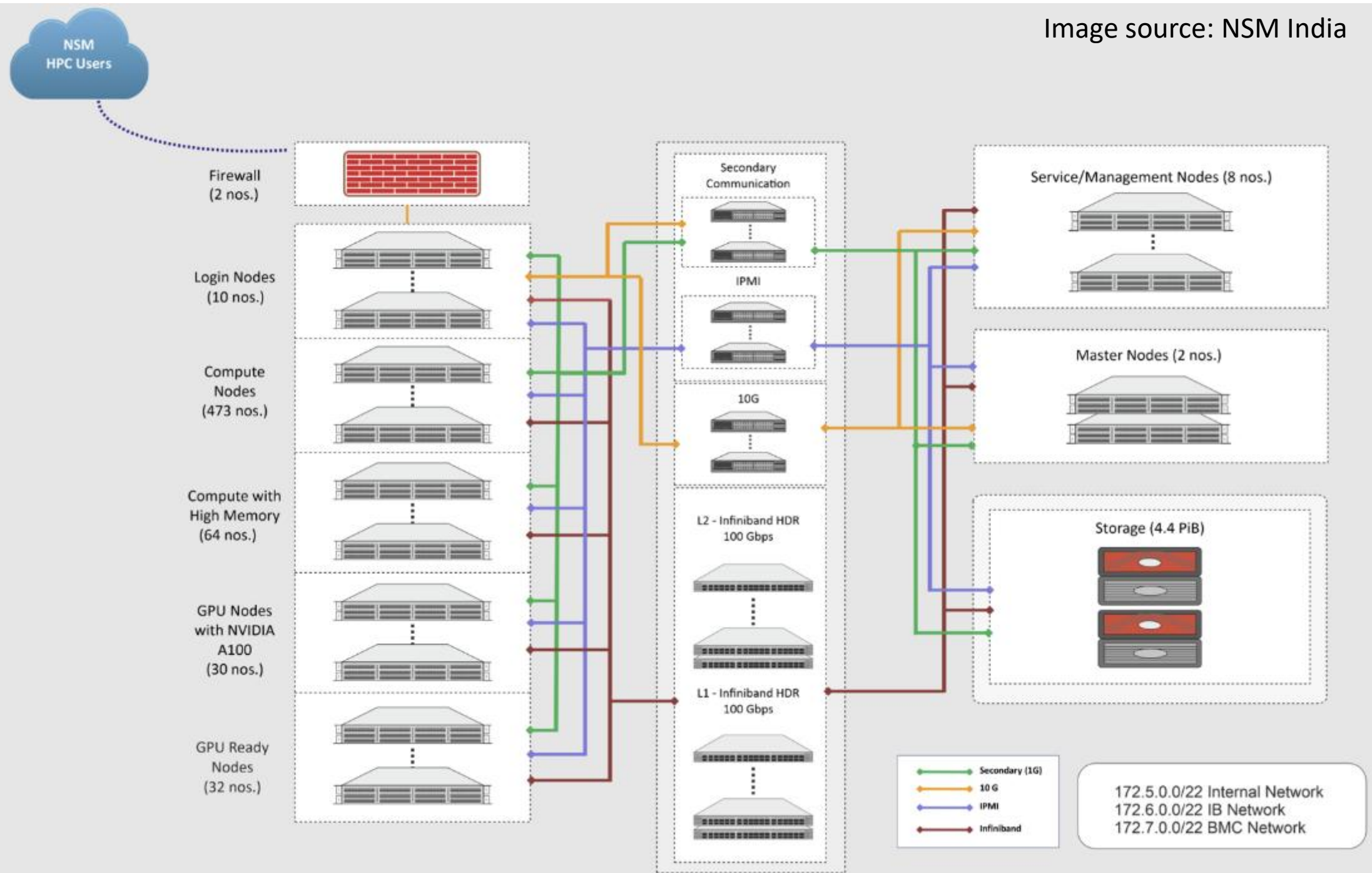
- [Performance analysis of k-ary n-cube interconnection networks](#)
- [Implementing the MPI Process Topology Mechanism](#)
- [Automated mapping of regular communication graphs on mesh interconnects](#)
- [The PERCS High-Performance Interconnect](#)
- [Reconfigurable Hybrid Interconnection for Static and Dynamic Scientific Applications](#)

PARAM Rudra C-DAC Delhi

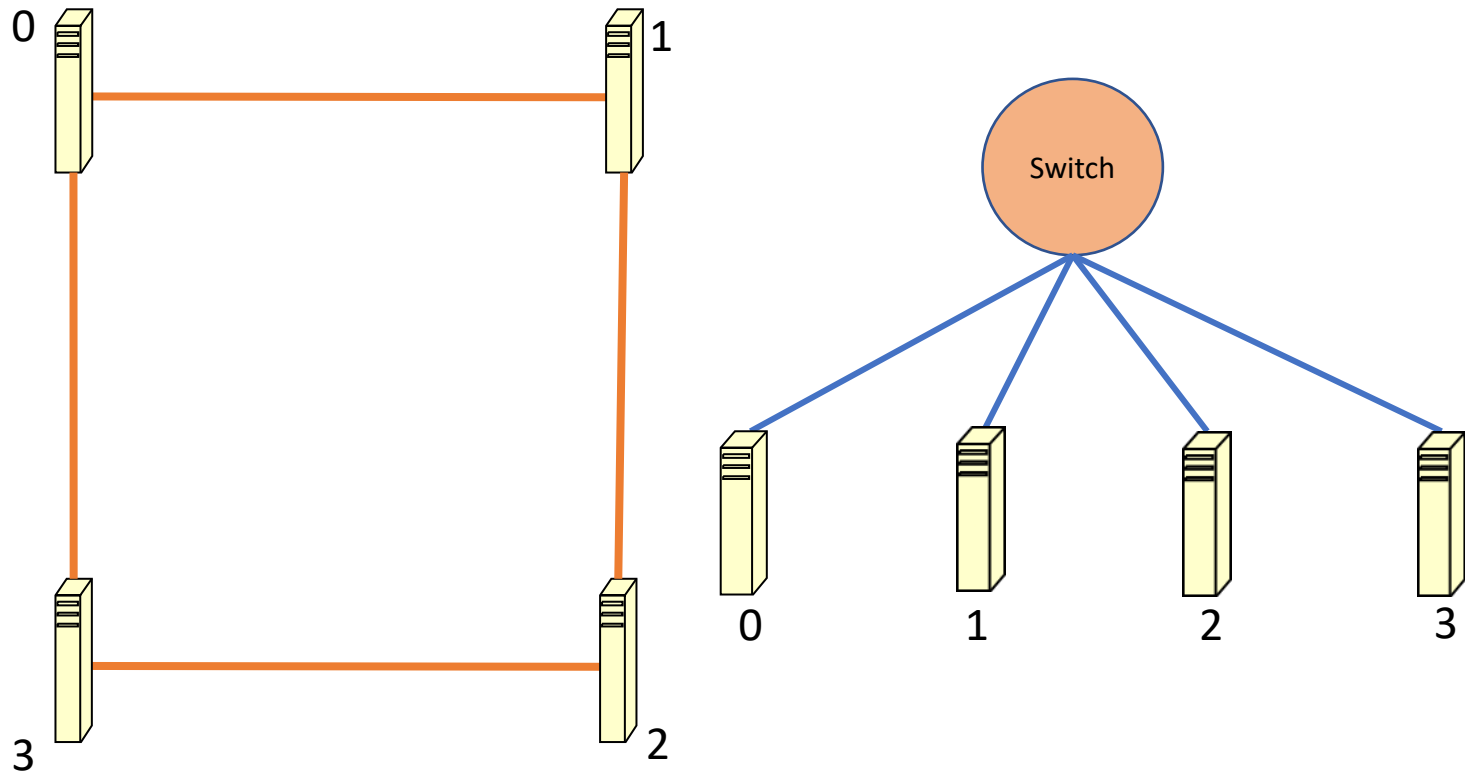
Image source: NSM India

- 100 Gbps Mellanox Infiniband interconnect (fat-tree topology)





Effect of Network Topology on Communication Time



Communication pattern: 0 $\rightarrow$ 2, 1 $\rightarrow$ 3 (N bytes)
Latency: L, Link bandwidth: b bytes/sec
Communication time comparison (ideal): ?

The topology on the left may incur slightly higher congestion for the given communication pattern. Reason: 0 $\rightarrow$ 2 communication path overlaps partially with 1 $\rightarrow$ 3 communication path in the mesh. In case of the right side topology, the switch has multiple ports to which each compute node is connected, and the communication paths may not necessarily overlap over a particular link (in contrast to the case of a mesh).

Effect of Node Allocation on Communication Time

HPC 2010

HPC2010 is part of high performance computation facility at Indian Institute of Technology Kanpur and funded by the Department of Science and Technology (DST), Govt. of India,. This machine has 376 nodes of which 368 are compute nodes. At the time of its installation, this machine had a rank of 369 in the top 500 list published in www.top500.org in June 2010.

In the initial ratings, it had a Rpeak of 34.50 Terra-Flops and Rmax of 29.01 Terra-Flops.

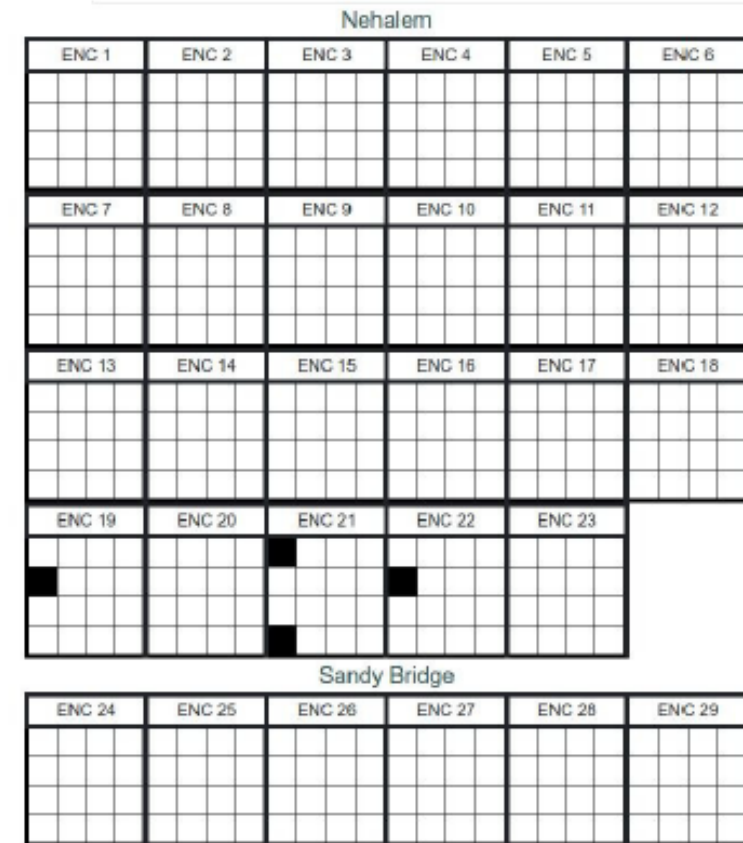
System Details:

It is based on Intel Xeon X5570 2.93 GHz 2 CPU-Nehalem (8-cores per node) with 48 GB of RAM per node. This cluster was later augmented with 96 nodes of Intel Xeon ES-2670 2.6 GHz (16-cores per node) with 64 GB of RAM per node that add an additional theoretical 31 Terra-Flops to the above 2010 cluster. PBS Pro is the scheduler of choice. Though it has FDR Infiniband cards it is connected to the QDR Infiniband fabric seamlessly through fat-tree topology.

The nodes are connected by Qlogic QDR Infiniband federated switches that can provide 40 Gbps of throughput. It also has 100 Terra-Bytes of storage with an aggregate performance of around 5 Gbps on write performance.

For more information download the data sheet.

<https://www.hpc.iitk.ac.in>



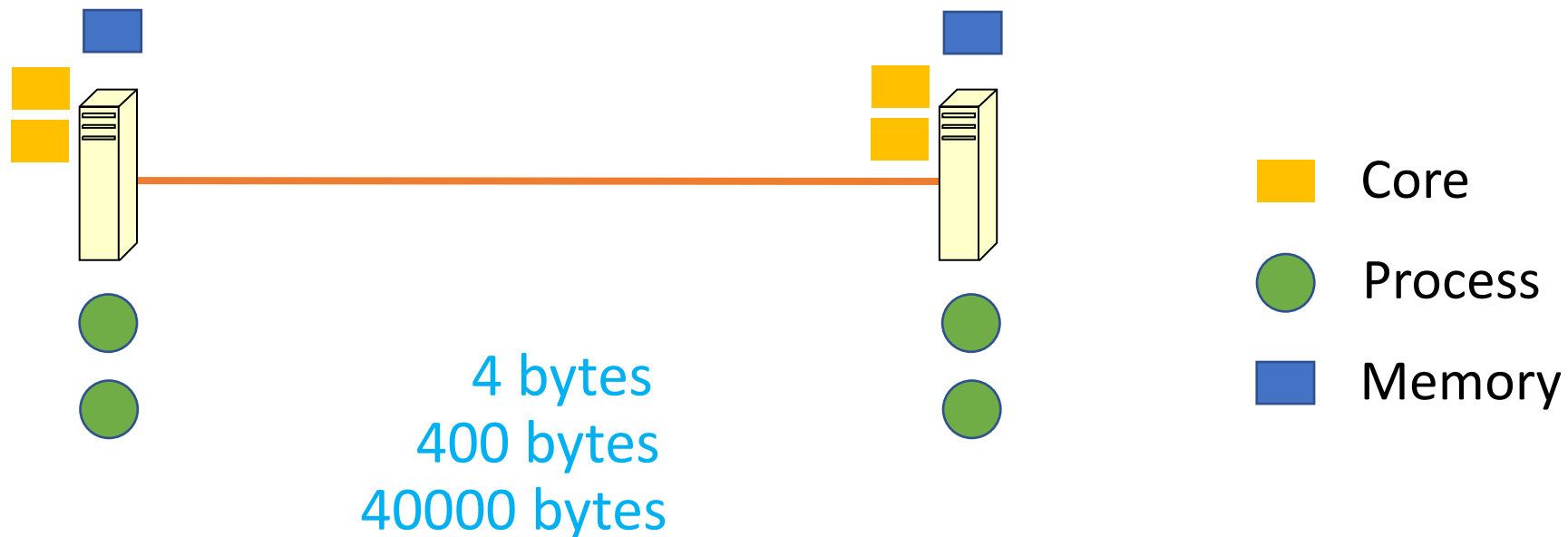
Source: A Pal et al.

H.W: Assume that 4 ranks are placed on the 4 different enclosures (switches). 0 on 19, 1 and 2 on 21, and 3 on 22. If the communication pattern is all-to-all, what optimization can you think of, considering the above process placement and job allocation.

Communication cost

MPI Communication

We'd like communication time to be negligible!
However, communication is often the bottleneck in parallel programs.

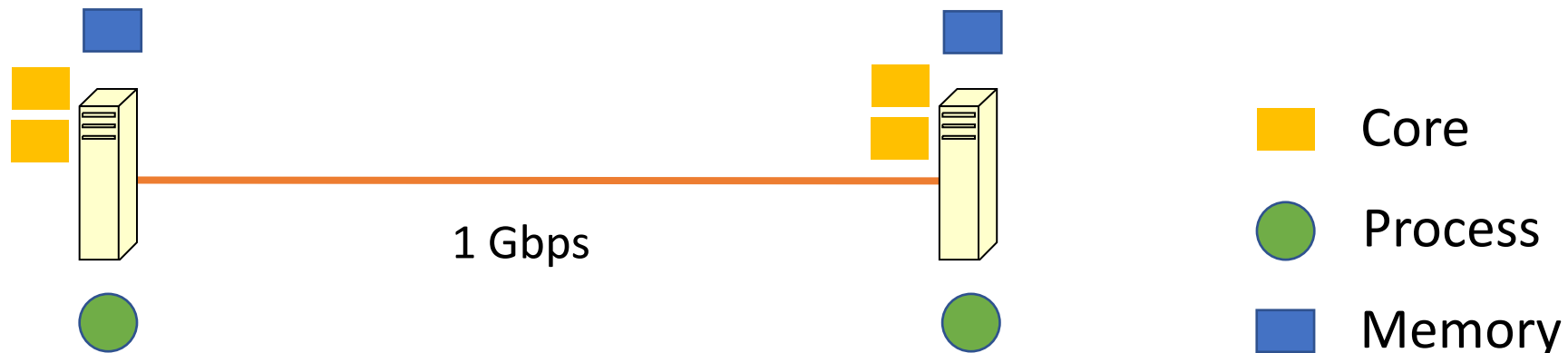


Homework: Time the communications (Use P2rutor)

Bandwidth/Throughput

- Amount of data that can be transmitted in a certain amount of time
- Unit?
- Example: 200 bits in 2 ms

→ Achieved bandwidth = 10^5 bps



Homework: Write a P2P code, run with $np=2,4,6,..$ and $ppn=1,2,3,..$ on Prutor.
Expected vs. observed?

MPI Overhead

Eager

700000 bytes (1000 times) in 6.02 sec → Effective bandwidth achieved = 111.2 MB/s

Rendezvous

700000 bytes (1000 times) in 6.25 sec → Effective bandwidth achieved = 106.8 MB/s

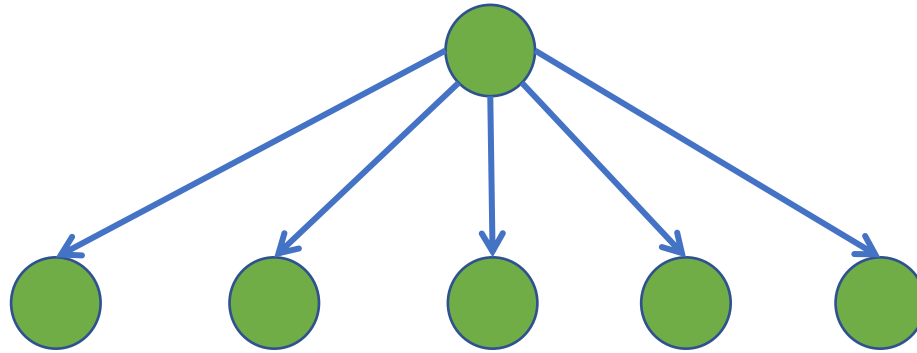
```
-----  
Client connecting to csews2, TCP port 5001  
TCP window size: 85.0 KByte (default)  
-----
```

```
[ 3] local 172.27.19.44 port 57146 connected with 172.27.19.2 port 5001  
[ ID] Interval      Transfer      Bandwidth  
[ 3]  0.0-10.0 sec  1.01 GBytes   870 Mbits/sec
```

Bits/Bytes/K/M

- b – bit
- B – Byte (8 bits)
- Kb – Kilobits (1024 bits)
- KB – Kilobytes (1024 bytes)
- Mb – Megabits (1024 bits)
- MB – Megabytes (1024 bytes)

Timing Analysis for 1-to-many Sends



I Communication volume = 10000 integers to 29 ranks in 0.02 s

➔ Throughput/bandwidth achieved = $10000 * 4 * 29 \text{ bytes} / 0.02 \text{ s} = 55.3 \text{ Mbps}$

II Communication volume = 1000000 integers to 29 ranks in 0.8 s

➔ Throughput/bandwidth achieved = $1000000 * 4 * 29 \text{ bytes} / 0.8 \text{ s} = 138.3 \text{ Mbps}$

Note: This is not the network bandwidth